

Lecture 03 - Describing and Summarizing Data

This lecture introduces descriptive statistics as the first disciplined step in statistical work. Before we can model, test, forecast, or make decisions from data, we need to understand what the data represent and how their main features can be summarized. Descriptive statistics is the set of tools used to organize raw observations, reduce them to informative summaries, visualize their structure, and interpret the results in context.

The lecture has three connected parts. The first part explains the vocabulary of datasets and statistics: population, sample, observation, variable, statistic, parameter, estimator, estimate, and type of data. The second part introduces the most common numerical summaries: measures of location, measures of dispersion, percentiles, and the distinction between population and sample variance conventions. The third part applies the same descriptive logic to finance, where raw price data are usually transformed into returns before we summarize performance and risk.

1. Why Descriptive Statistics Comes First

Raw data rarely tell a story by themselves. A dataset may contain hundreds or thousands of prices, invoices, customers, days, or transactions. Looking at the raw list is usually not enough to answer a question. Descriptive statistics turns that list into something interpretable.

Summarization reduces many observations to a smaller number of meaningful quantities, such as an average, a proportion, a standard deviation, or a percentile. Visualization turns patterns into forms that the eye can recognize, such as bar charts, histograms, line charts, and scatterplots. Interpretation connects the numerical or visual summary back to the question we care about.

This last step is essential. A number is not useful simply because it has been computed. It becomes useful when we know what it describes, where it comes from, and what question it helps answer. For example, the statement “the standard deviation of daily returns over the last three years is 1.2 percent per day” is meaningful because it identifies a variable, a sample period, a unit, and an interpretation: the number describes day-to-day volatility in that historical sample.

Whenever you encounter a statistic, ask four questions:

1. Which variable does it describe?
2. Which sample was used?
3. Over what period was the sample observed?
4. What question is the statistic meant to answer?

Those questions are simple, but they prevent many common mistakes. A statistic without context can easily be misread.

2. The Structure of a Dataset

A dataset is built from a few fundamental ideas.

The population is the complete group, process, or set of outcomes that we ultimately care about. In finance, the population might be the return-generating process of an asset, all future trading days, all firms in a market, or all loans of a certain type. The population is often not fully observable.

The sample is the part of the population or process that we actually observe. If we use daily Apple prices from 2012 to 2021, that historical period is the sample. It is not the entire price process of Apple forever. Most statistical work uses samples to learn about something broader.

An observation is one unit in the dataset. It could be one customer, one invoice, one transaction, one trading day, one firm, or one firm-year. A variable is a characteristic measured for each observation, such as revenue, return, age, industry, satisfaction level, or payment method.

In a standard data table, rows represent observations and columns represent variables. This structure matters because it tells us what can be summarized. We usually summarize one variable across many observations, or study relationships between variables measured on the same observations.

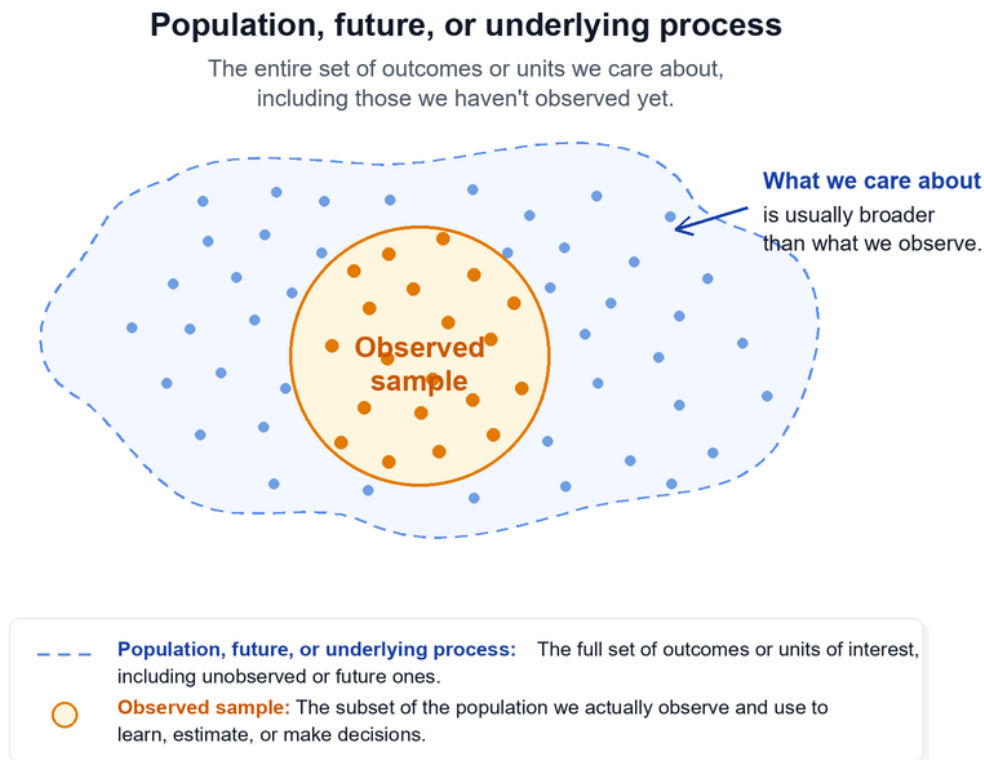


Figure 1: Observed sample inside a broader population process

Figure 1. An observed sample sits inside a broader population or process.

The figure emphasizes the distinction between what we observe and what we care about. The observed sample is the visible part of a larger population, future, or underlying process. Descriptive statistics begins with the observed sample, but careful interpretation remembers that the sample is only part of the bigger object of interest.

3. Statistic, Parameter, Estimator, and Estimate

Four terms are closely related but should not be confused.

A parameter is a numerical characteristic of the population. The true expected return of an asset, the true volatility of a return-generating process, or the true default probability of a loan portfolio are parameters. In practice, parameters are usually unknown.

A statistic is any numerical summary computed from a sample. The sample mean, sample median, sample standard deviation, sample minimum, sample maximum, and sample correlation are all statistics.

An estimator is the rule or formula used to estimate an unknown parameter. For example, the sample mean is an estimator of the population mean. The estimator is the method, not the numerical result.

An estimate is the number obtained when the estimator is applied to one particular sample. If a sample of monthly returns gives an average of 0.7 percent per month, then 0.7 percent is an estimate of the population mean monthly return.

Figure 2. The relationship between parameter, sample, estimator, and estimate.

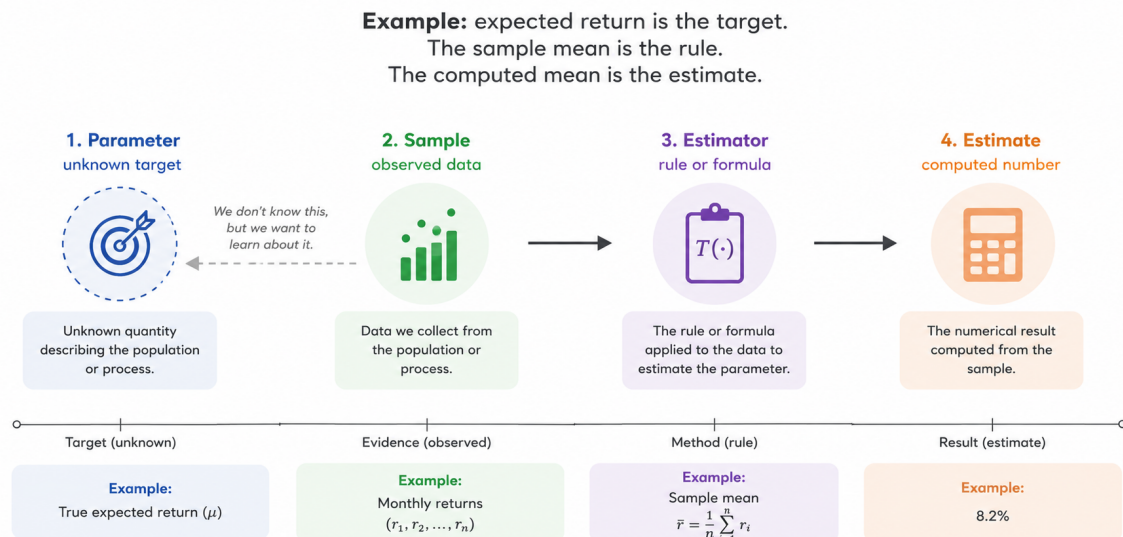


Figure 2: Relationship between parameter, sample, estimator, and estimate

This distinction is central to statistical thinking. A different sample would generally produce a different estimate, even if we use the same estimator. The estimate is therefore a sample-based number, not a guaranteed truth about the full population.

4. Types of Data

Before computing statistics, we need to identify the type of variable. In programming, a data type tells the computer how a value is stored. In statistics, a type of data tells us how the variable should be summarized, visualized, and analyzed. These two ideas are related, but they are not the same.

A nominal categorical variable contains labels with no natural order. Industry, country, and payment method are examples. We can count categories and identify the most frequent category, but taking an average of industry labels has no meaning.

An ordinal categorical variable contains categories with a meaningful order. Satisfaction levels such as low, medium, good, and excellent are ordinal. The order matters, but the distance between categories is not necessarily numerical.

A binary variable has two possible categories, such as default versus no default, yes versus no, or credit card versus no credit card. Binary variables are often stored as 0 and 1. When coded this way, the average can be interpreted as a proportion, but the variable is still conceptually categorical.

A discrete numerical variable is a count. Number of items bought, number of customers, and number of defaults are examples. These variables usually take integer values.

A continuous numerical variable is a measurement on a scale. Returns, prices, heights, weights, and times are examples. In practice, continuous values are recorded with finite precision, but conceptually they can vary over an interval.

Figure 3. Main statistical types of variables and their common Python storage types.

The figure connects statistical variable types with common Python storage types. Strings and booleans often store categorical variables; integers often store discrete numerical variables; floats often store continuous variables. But the correspondence is only approximate. An invoice ID may be stored as an integer, but it is not meaningful to average invoice IDs. A 0/1 variable may be stored as an integer, but its main summaries are counts, proportions, and category frequencies.

The type of data determines the type of summary that is meaningful. For nominal categories, frequencies and proportions are natural. For ordinal categories, frequencies and ordered compar-

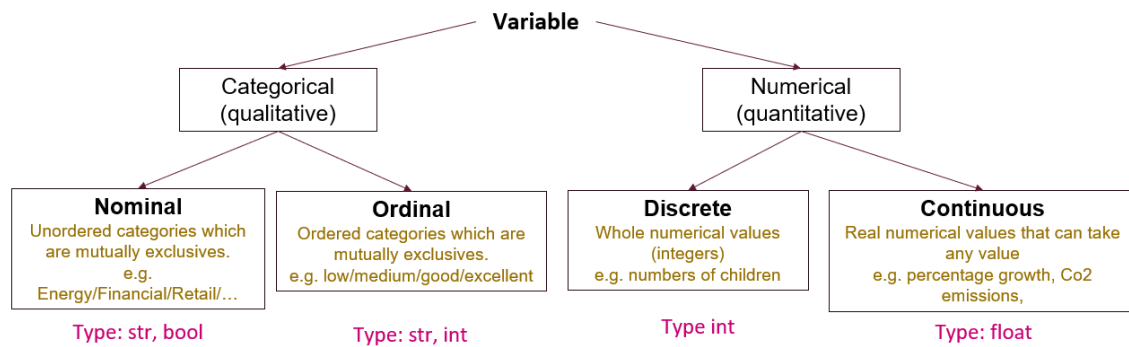


Figure 3: Main statistical types of variables and common Python storage types

isons are natural. For numerical variables, means, standard deviations, percentiles, and histograms become meaningful.

5. From Raw Data to Frequency Tables

Many descriptive summaries begin by counting.

For categorical data, a frequency table reports how many times each category appears. If a satisfaction variable contains the labels poor, average, good, and very good, the frequency table counts how many observations fall in each category. A relative frequency table divides each count by the total sample size, so that the entries become proportions or percentages.

For discrete numerical data, the same idea applies. If the number of items bought takes values 0, 1, 2, 3, 4, and 5, we can count how many observations have each value.

Frequency tables are useful because they compress raw data into an interpretable form. A raw list of 100 category labels is hard to read. A table with category counts is much easier to interpret. A bar chart then turns those counts into a visual comparison.

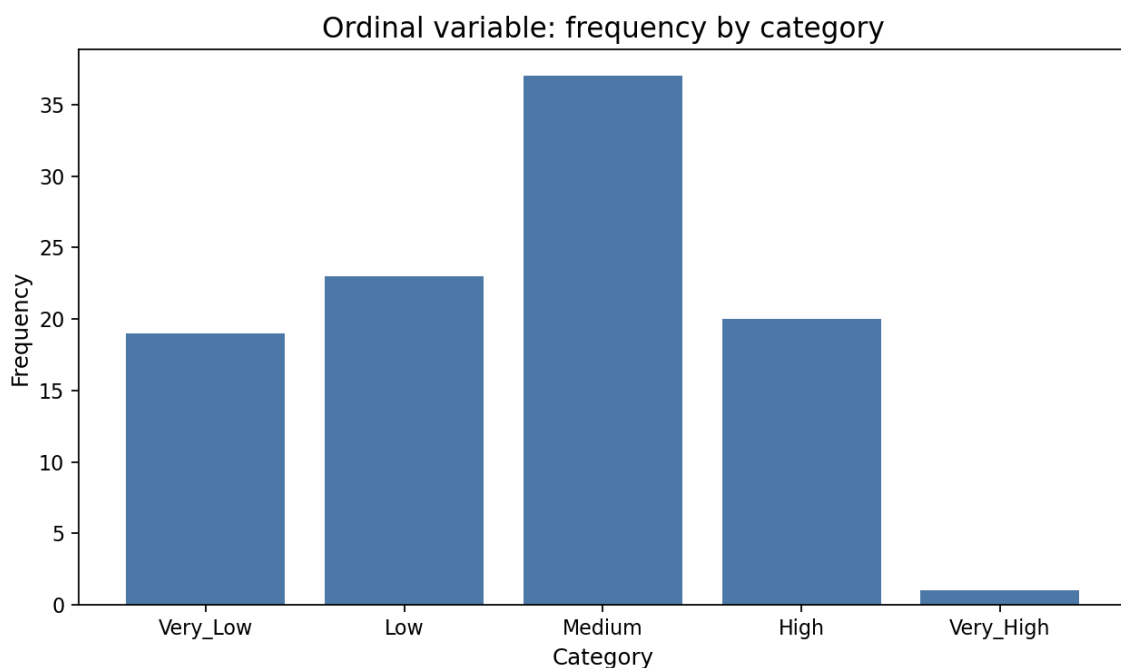


Figure 4: Bar chart for an ordinal categorical variable

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The bar chart above is a frequency display. Each bar corresponds to a category, and the height of the bar is the number of observations in that category. For ordinal categories, the order of the categories should usually be preserved along the horizontal axis, because the order itself carries information.

For continuous numerical variables, counting exact repeated values is usually not useful. A dataset of 500 continuous measurements may contain many distinct values, each appearing only once. The solution is to group values into intervals, called bins, and count how many observations fall in each bin. This grouped frequency table is the basis of a histogram.

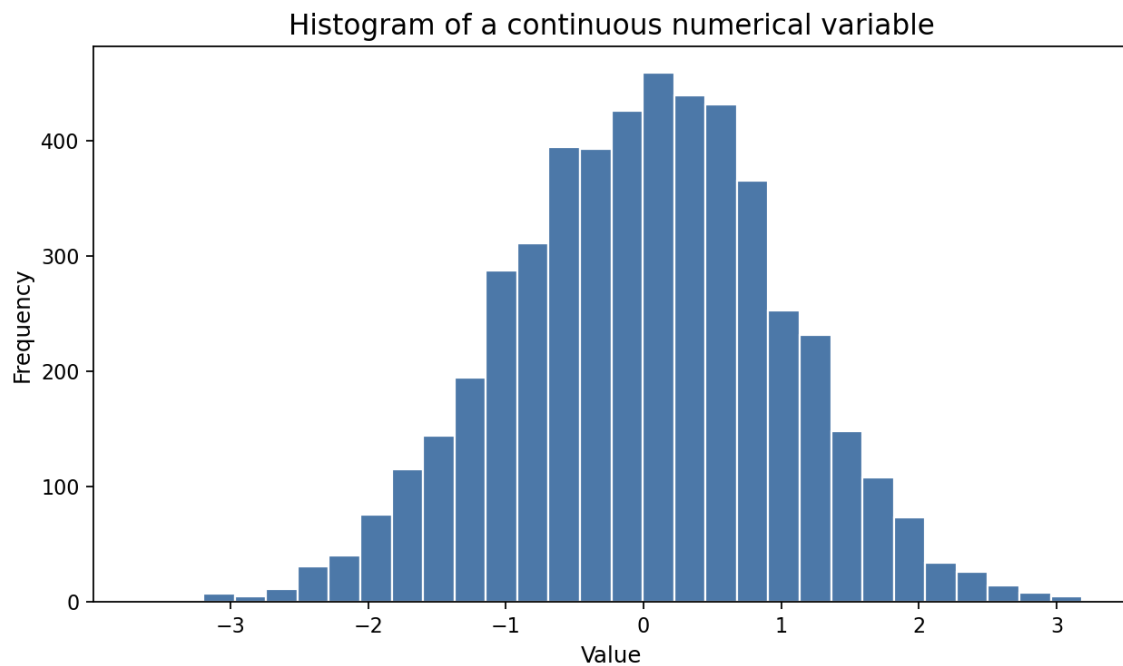


Figure 5: Histogram of a continuous numerical variable

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A histogram is the graphical representation of a grouped frequency table for a numerical variable. The horizontal axis represents numerical intervals, and the vertical axis represents frequency. Unlike a bar chart for categories, the order and spacing of the bins matter. The bars show how the observations are distributed along the numerical scale.

The choice of bin width matters. Wide bins produce a smoother but less detailed summary. Narrow bins preserve more detail but may make the distribution look noisy. A histogram should therefore be read as a summary of the distribution, not as a perfect representation of every individual observation.

6. Choosing a Visualization

Different variables call for different visualizations.

For nominal and ordinal categorical data, bar charts are usually appropriate because they compare category frequencies. Pie charts can show category shares, but they are often hard to read when there are many categories or when the differences between categories are small.

For a single continuous numerical variable, a histogram is usually the first graph to inspect. It shows the approximate center, the spread, the skewness (whether the distribution leans more to one side than the other), and possible extreme values of the distribution.

For values ordered over time, a line chart is often useful. A price series, an index level, or a time series of sales can be plotted against time to show movement and trend.

For two numerical variables measured on the same observations, a scatterplot is the natural first graph. It lets us see whether the variables tend to move together, whether the relationship is positive or negative, whether it is linear or nonlinear, and whether there are outliers or clusters.

The graph should follow the question. A chart is not automatically useful because it is attractive. It is useful when it helps answer a statistical or business question more clearly than a table alone.

7. Measures of Location

Once a variable is identified as numerical, one of the first questions is where the data are located. Measures of location describe a typical level or center of the data.

The arithmetic mean, often called the average, is the sum of all observations divided by the number of observations. For observations $x_1, \dots, x_n$, the sample mean is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

The mean uses every value in the dataset. This is an advantage because all observations contribute, but it also makes the mean sensitive to extreme observations.

The median is the middle value after the observations are sorted from smallest to largest. If the sample size is odd, the median is the value in the middle position. If the sample size is even, the median is usually computed as the average of the two middle values. The median depends on ordered position, not directly on the magnitude of every observation, which makes it more robust to extreme values.

The mode is the most frequently occurring value. It is useful for categorical variables as well as numerical variables. For continuous data, the exact mode of raw values is often less meaningful, so the idea is usually applied to a grouped distribution or histogram.

Consider the ordered dataset

1, 2, 4, 5, 5, 5, 6, 6, 8, 9, 10.

The mean is $61/11 \approx 5.55$, the median is 5, and the mode is 5. These measures are similar because the dataset is fairly balanced around its center.

Now add one extreme observation:

1, 2, 4, 5, 5, 5, 6, 6, 8, 9, 10, 35.

The mean becomes $96/12 = 8$, while the median becomes 5.5 and the mode remains 5. One unusually large value changes the mean substantially but has a much smaller effect on the median and no effect on the mode. This example illustrates why the mean is not always the best description of a typical value, especially when a dataset contains outliers.

Figure 6. Mean and median on an empirical distribution.

The histogram above shows a distribution with vertical reference lines for the mean and median. When a distribution is roughly symmetric, the two are often close. When a distribution is skewed or contains extreme values, the mean can be pulled toward the tail.

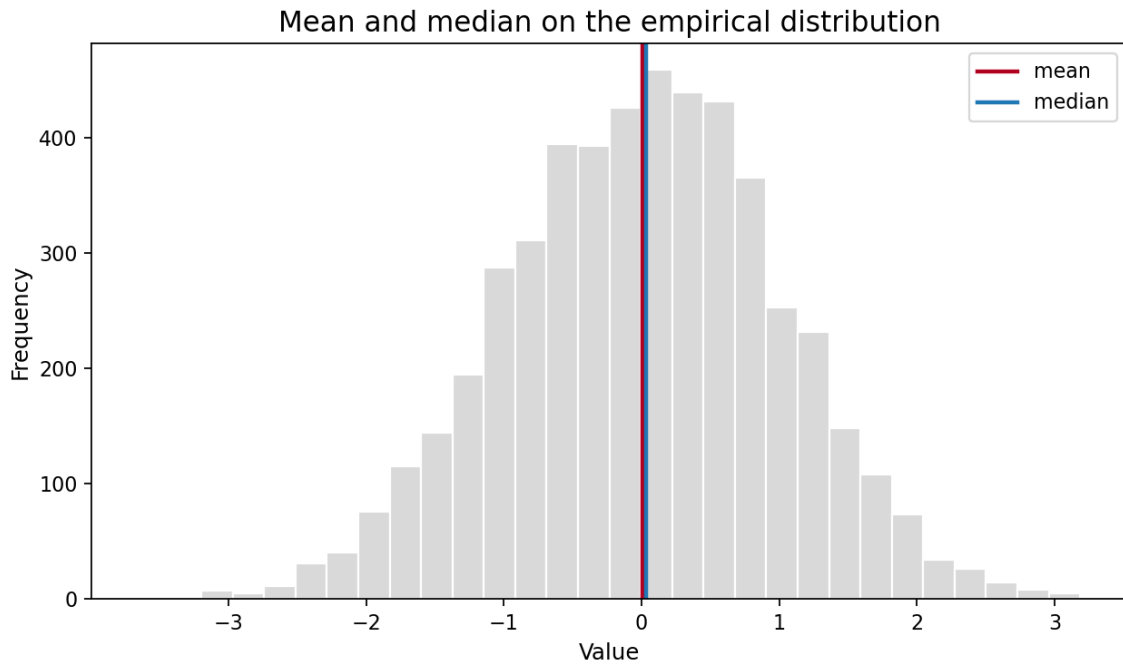


Figure 6: Mean and median on an empirical distribution

8. Mean from Tabulated Data

Sometimes we do not have the raw data in a list. Instead, the data may already be summarized in a frequency table. In that case, the mean is computed as a weighted average.

Suppose x_j is a possible value and f_j is its frequency. If there are k distinct values, the mean is

$$\bar{x} = \frac{\sum_{j=1}^k x_j f_j}{\sum_{j=1}^k f_j}.$$

The denominator is the total number of observations. The numerator counts each value as many times as it appears.

For example, suppose the number of items bought is tabulated as follows.

Items bought	Frequency	Contribution
1	6	6
2	14	28
3	21	63
4	12	48
5	7	35
Total	60	180

The mean is $180/60 = 3$. This is not the same as averaging the displayed values 1, 2, 3, 4, and 5, because those values do not appear equally often. The frequencies tell us how much weight each value has in the dataset.

9. Percentiles and Quartiles

Measures of location are not limited to the center. Percentiles describe position across the ordered dataset.

The p -th percentile is a value such that approximately p percent of the observations are at or below it. The exact computational rule can differ slightly across software packages, especially when the desired percentile falls between two observed values, but the interpretation is stable: percentiles describe ordered position.

Three percentiles are especially common:

Statistic	Percentile	Interpretation
Q_1	25th	About one quarter of observations are at or below this value.
Q_2	50th	The median. About half of observations are at or below this value.
Q_3	75th	About three quarters of observations are at or below this value.

Percentiles are useful because they describe more than the center. They show how values are distributed across the sample. The interquartile range,

$$IQR = Q_3 - Q_1,$$

measures the width of the middle 50 percent of the data. In finance, percentiles are often used to describe downside outcomes, tail observations, or thresholds.

In Python, percentiles can be computed with functions such as `np.percentile()` or with pandas methods such as `.quantile()`.

10. Measures of Dispersion

Location tells us where the data are centered. Dispersion tells us how spread out the observations are around that center. Dispersion is especially important in finance because it is closely related to risk.

Two datasets can have the same mean and very different dispersion. Consider the following two datasets:

$$A = \{8, 9, 10, 11, 12\}$$

and

$$B = \{0, 5, 10, 15, 20\}.$$

Both have mean 10, but dataset B is much more spread out. If these were investment returns, the same average return would not imply the same risk.

Figure 7. Two datasets with the same mean but different dispersion.

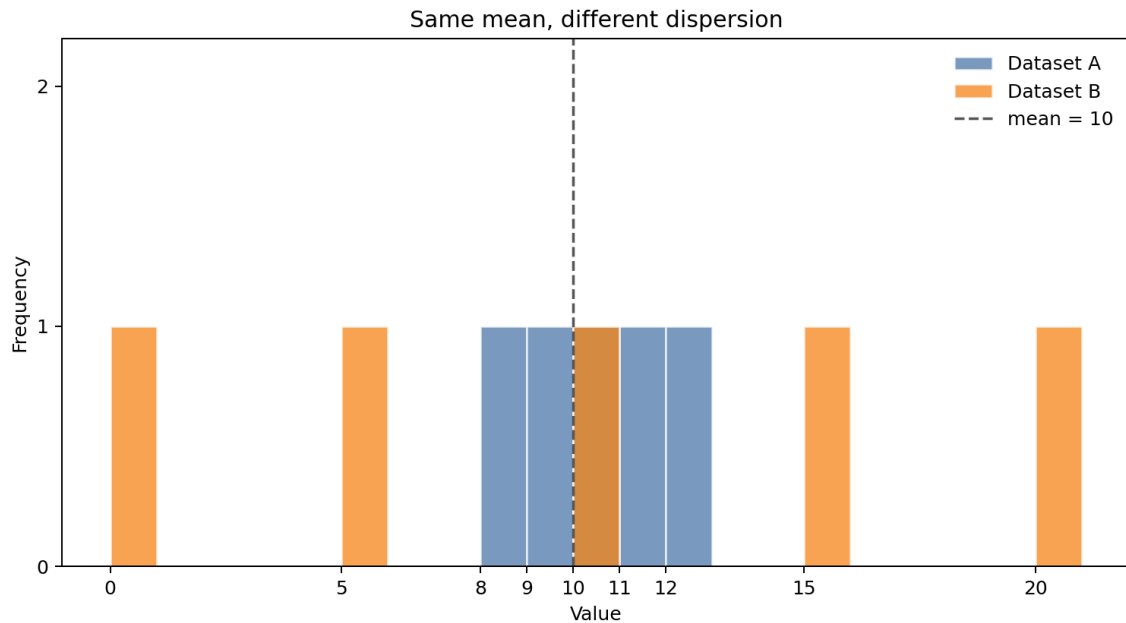


Figure 7: Two datasets with the same mean but different dispersion

The range is the simplest measure of dispersion:

$$Range = \max(x) - \min(x).$$

It is easy to interpret, but it depends only on the two most extreme observations. It ignores everything between the minimum and maximum.

The variance measures the average squared distance of observations from their mean. For a full population with mean μ , the population variance is

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2.$$

For a sample with mean $\bar{x}$, the usual sample variance is

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

The squared deviations ensure that positive and negative deviations do not cancel. Squaring also gives more weight to large deviations.

The standard deviation is the square root of the variance:

$$s = \sqrt{s^2}.$$

The standard deviation is easier to interpret than the variance because it is expressed in the original unit of the variable. If returns are measured in percent per day, the standard deviation is also measured in percent per day. If invoice amounts are measured in euros, the standard deviation is measured in euros.

The coefficient of variation compares dispersion with the mean:

$$CV = \frac{s}{\bar{x}}.$$

This can be useful when comparing variability across variables with different scales, but it should be used carefully. It is not meaningful when the mean is close to zero, and it is not appropriate when the mean itself is not a meaningful benchmark.

11. The n Versus $n - 1$ Convention

Variance and standard deviation have two common denominators.

If the data are treated as the full population we want to summarize, it is natural to divide by N :

$$\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2.$$

If the data are treated as a sample used to estimate a larger population variance, the usual sample variance divides by $n - 1$:

$$\frac{1}{n - 1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

The version with $n - 1$ is slightly larger because $n - 1$ is smaller than n . The difference matters most for small samples. For large samples, the difference becomes small.

This convention also explains why Python functions may give slightly different answers. By default, `numpy.var()` and `numpy.std()` use the n denominator, while `pandas .var()` and `.std()` use the $n - 1$ denominator by default. The parameter `ddof` controls this convention: `ddof=0` divides by n , while `ddof=1` divides by $n - 1$.

The important point is not to memorize every library default. The important point is to report and interpret the convention used.

12. From Prices to Returns

In finance, raw data are often prices:

$$P_0, P_1, P_2, \dots, P_T.$$

Prices are important, but they are not always the best object for statistical comparison. A change of 1 euro means something different for an asset priced at 10 euros than for an asset priced at 100 euros. Returns solve this problem by measuring the change relative to the starting price.

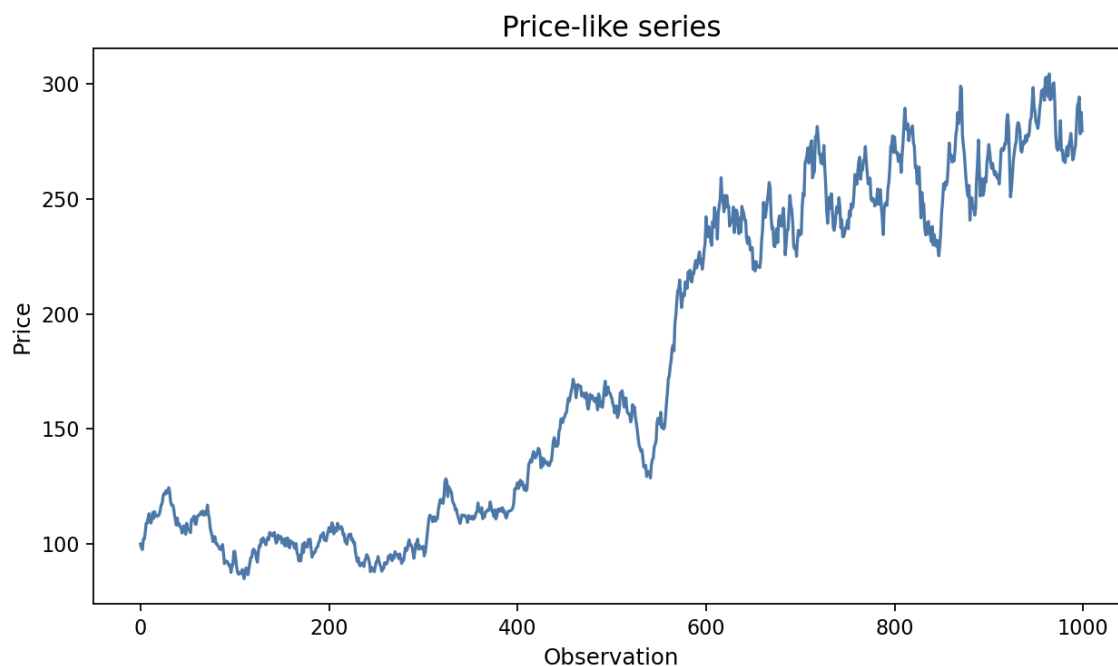
Figure 8. A price-like (level) series.

A price series is a level series. It shows how the asset value evolves over time, but its scale depends on the asset price level. To compare performance and risk across assets, we usually transform prices into returns.

The simple return from period $t - 1$ to period t is

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} - 1.$$

A simple return of 0.02 means that the asset gained 2 percent over the period. A simple return of -0.01 means that it lost 1 percent.

**Figure 8:** Price-like series

The log return is

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right).$$

Because

$$\frac{P_t}{P_{t-1}} = 1 + R_t,$$

we can also write

$$r_t = \ln(1 + R_t).$$

For small returns,

$$\ln(1 + R_t) \approx R_t.$$

This approximation explains why simple returns and log returns are often numerically close at daily frequency. They are not identical, however, and they combine differently over time.

13. Compounding Simple Returns

Simple returns compound multiplicatively.

Suppose R_t is the simple return from $t-1$ to t , and R_{t+1} is the simple return from t to $t+1$. The two-period simple return is not exactly $R_t + R_{t+1}$, because the second return applies to the value after the first return has already occurred.

A simple return corresponds to a growth factor $1 + R$. Therefore, the two-period growth factor is

$$1 + R_{t-1 \rightarrow t+1} = (1 + R_t)(1 + R_{t+1}).$$

So the two-period return is

$$R_{t-1 \rightarrow t+1} = (1 + R_t)(1 + R_{t+1}) - 1.$$

More generally, over T periods,

$$1 + R_{0 \rightarrow T} = \prod_{t=1}^T (1 + R_t),$$

and therefore

$$R_{0 \rightarrow T} = \prod_{t=1}^T (1 + R_t) - 1.$$

This is the logic behind compounding. If an asset gains 10 percent and then loses 10 percent, the total return is not zero:

$$(1.10)(0.90) - 1 = -0.01.$$

The asset is down 1 percent because the loss is applied to a larger base after the first gain.

14. Compounding Log Returns

Log returns add over time.

Using the same two periods, define

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right)$$

and

$$r_{t+1} = \ln \left(\frac{P_{t+1}}{P_t} \right).$$

The full-period log return from $t - 1$ to $t + 1$ is

$$r_{t-1 \rightarrow t+1} = \ln \left(\frac{P_{t+1}}{P_{t-1}} \right).$$

Insert the intermediate price P_t by multiplying and dividing by P_t :

$$\frac{P_{t+1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} \times \frac{P_{t+1}}{P_t}.$$

Then use the logarithm rule $\ln(ab) = \ln(a) + \ln(b)$:

$$r_{t-1 \rightarrow t+1} = \ln \left(\frac{P_t}{P_{t-1}} \right) + \ln \left(\frac{P_{t+1}}{P_t} \right) = r_t + r_{t+1}.$$

This additivity is the main practical advantage of log returns. When we move through time, log returns add cleanly. Simple returns are more intuitive for one-period percentage gains and losses; log returns are often more convenient for multi-period calculations and mathematical work.

15. Describing Return Distributions

Once returns have been computed, they can be summarized like any other numerical variable. We can compute the sample mean return, median return, standard deviation, quartiles, minimum, and maximum. We can also draw a histogram.

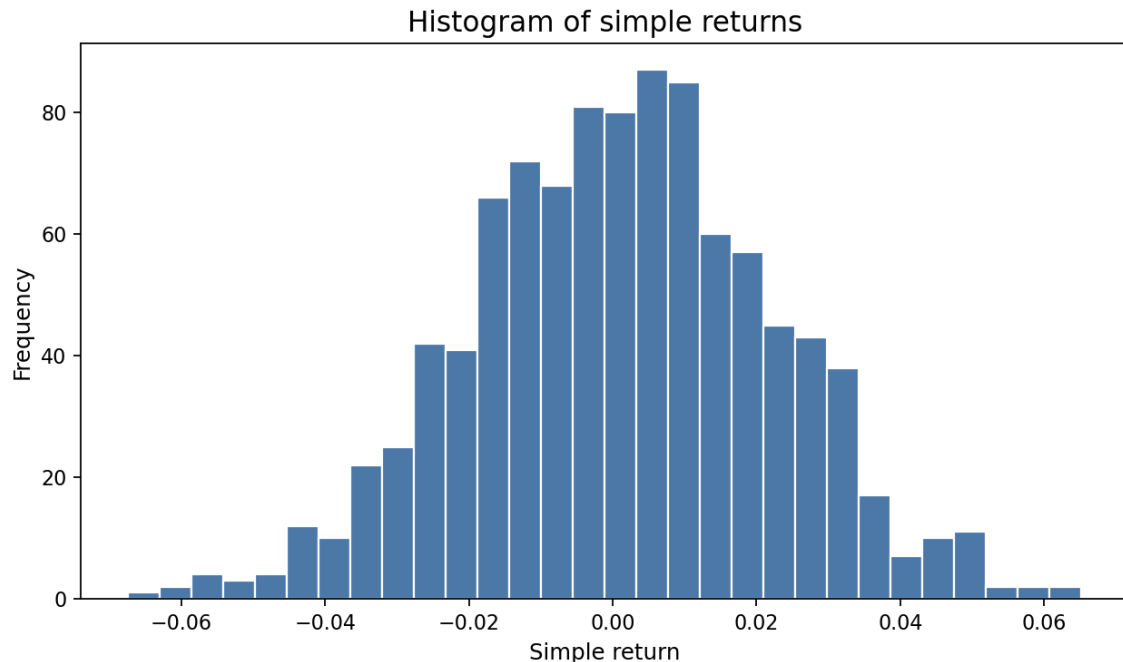


Figure 9: Histogram of simple returns

Figure 9. Histogram of simple returns.

The histogram of returns shows the empirical distribution of historical returns. The center describes the typical daily movement. The spread describes volatility. The tails show unusually large positive or negative observations. A histogram is often more informative than the mean alone, because financial return distributions may have extreme values and may not be perfectly symmetric.

When reporting descriptive statistics for returns, always specify the asset, sample period, return definition, and frequency. For example:

In this historical sample of daily simple returns, the mean daily return is 0.115 percent and the daily standard deviation is 1.793 percent. These are descriptive statistics for the sample, not forecasts of future performance.

That final sentence is not a formality. Descriptive statistics summarize observed data. They do not by themselves prove what will happen in the future.

16. Annualizing Average Returns

Daily returns are often difficult to interpret directly. Finance commonly reports performance and risk on an annual scale, so we often annualize returns.

Let $\bar{R}_{daily}$ be the average daily simple return. If there are approximately 252 trading days in a year, the compounded annualized simple return is

$$\bar{R}_{annual} = (1 + \bar{R}_{daily})^{252} - 1.$$

More generally, if there are m periods per year,

$$\bar{R}_{annual} = (1 + \bar{R}_{period})^m - 1.$$

This formula compounds the average period return repeatedly over the year. It is exact for the hypothetical situation in which the average period return is earned every period.

For small period returns, a common linear approximation is

$$\bar{R}_{annual} \approx m\bar{R}_{period}.$$

The approximation is often close for small daily returns, but it ignores compounding.

For average log returns, annualization is simpler. If $\bar{r}_{daily}$ is the average daily log return, then

$$\bar{r}_{annual} = 252\bar{r}_{daily},$$

or, more generally,

$$\bar{r}_{annual} = m\bar{r}_{period}.$$

This works because log returns are additive over time. If we want to convert an annualized log return back into a simple return, we use

$$\exp(\bar{r}_{annual}) - 1.$$

17. Annualizing Variance and Volatility

Risk is often summarized by variance or standard deviation. If returns are observed daily, we can compute daily variance and daily standard deviation, but we often want an annual measure.

Under the usual square-root-of-time convention, variance scales linearly with time:

$$s_{annual}^2 = 252s_{daily}^2.$$

More generally,

$$s_{annual}^2 = ms_{period}^2.$$

Since standard deviation is the square root of variance, volatility scales with the square root of time:

$$s_{annual} = \sqrt{252} s_{daily},$$

or, more generally,

$$s_{annual} = \sqrt{m} s_{period}.$$

This rule is widely used, but it relies on assumptions. It works best when returns are measured at the same frequency, have reasonably stable variance, and are not strongly correlated over time. If returns are strongly autocorrelated, covariance terms matter, and the simple scaling rule becomes less reliable.

An example makes the scale clear. Suppose the average daily simple return is 0.05 percent, or 0.0005, and daily volatility is 1.2 percent, or 0.012. With $m = 252$ trading days:

$$\bar{R}_{annual} = (1 + 0.0005)^{252} - 1 \approx 0.1342 = 13.42\%.$$

Annualized volatility is

$$s_{annual} = \sqrt{252} \times 0.012 \approx 0.1905 = 19.05\%.$$

The linear approximation to the annualized return is

$$252 \times 0.0005 = 0.126 = 12.60\%.$$

The approximation is close, but it is lower than the compounded value by about 0.82 percentage points. The difference is small in this example because the daily return is small, but the gap becomes larger when returns per period are larger or when the number of compounding periods increases.

18. How to Report Descriptive Statistics Carefully

A good descriptive statement does more than list numbers. It answers the four questions from Section 1 — which variable, which sample, over what period, and what question the number addresses — and adds two practical details that make the result reproducible: the frequency at which the data are observed and the calculation method used.

For a customer dataset, a careful statement might say:

In this sample of 60 invoices, the mean number of items bought is 3. The calculation uses a frequency table, so each item count is weighted by the number of invoices with that count.

For a financial return dataset, a careful statement might say:

In this sample of Apple adjusted closing prices from 2012-01-03 to 2021-10-01, daily simple returns have a sample mean of 0.115 percent and a sample standard deviation of 1.793 percent. The annualized volatility using the square-root-of-time rule is approximately 28.45 percent. These numbers describe this historical sample and should not be interpreted as guaranteed future performance.

The exact numerical values will depend on the data file and calculation convention, especially for the standard deviation denominator and annualization assumptions.

The discipline is the same throughout the lecture. First, identify the variable and its type. Second, choose summaries and graphs that are meaningful for that type. Third, compute the statistics carefully. Fourth, interpret them in context, remembering the distinction between the sample we observe and the broader population or process we often care about.